

Dawning4000A

A 10TFlops Linux Cluster

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**Institute of Computing Technology
Chinese Academy of Sciences**

Dec 3th, 2003. Cluster2003

Background

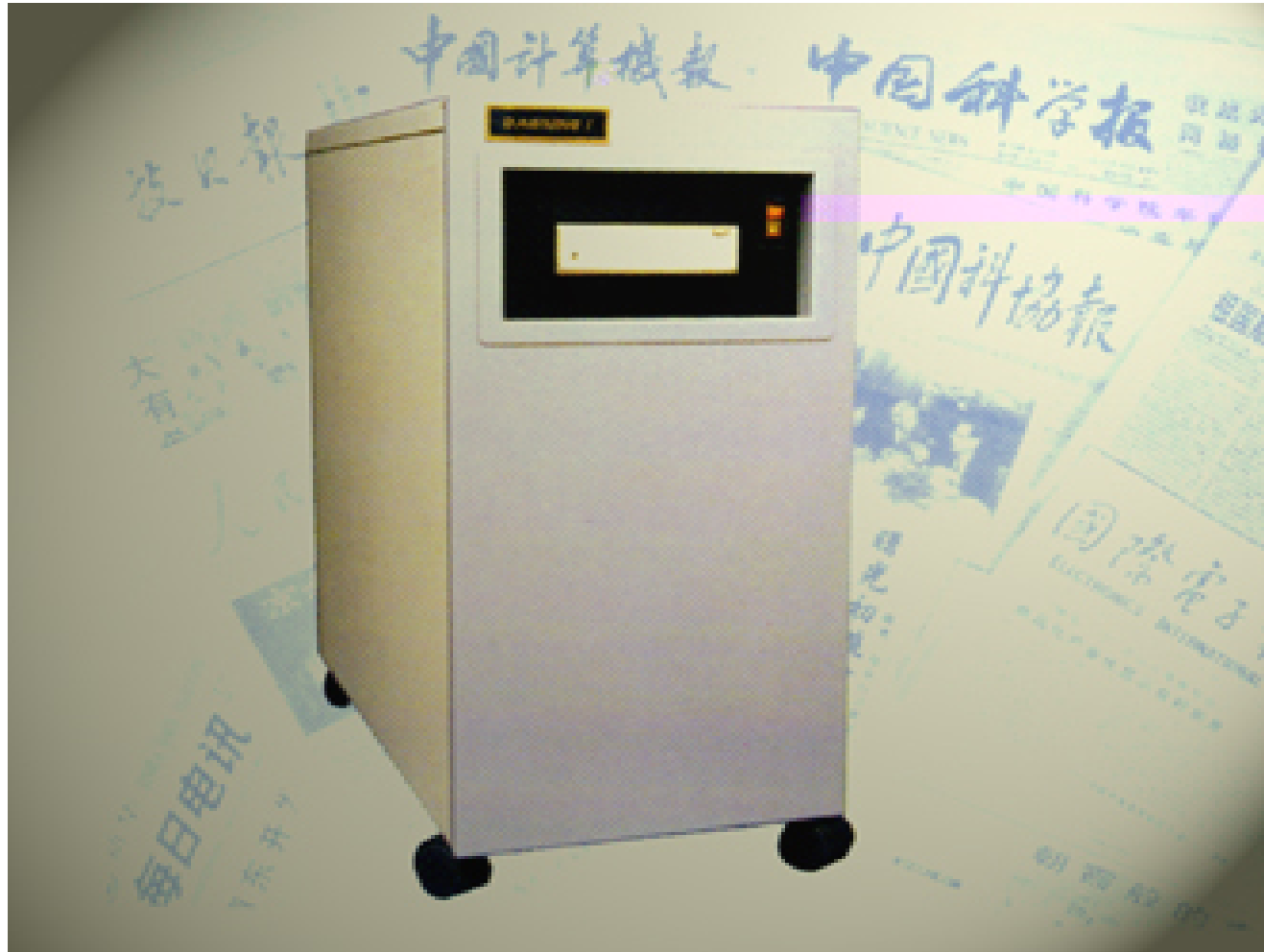


中国科学院计算技术研究所
Institute of Computing Technology, Chinese Academy of Sciences



Established in 1956

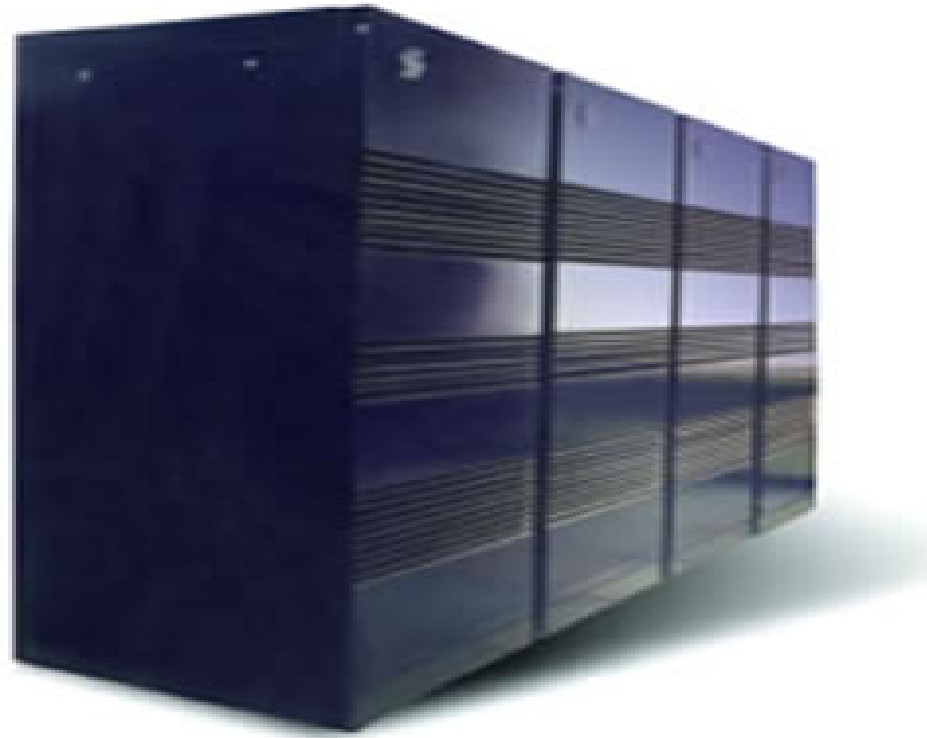
First computing research institute in China



Dawning-1
First SMP Computer in China
1993



Dawning1000
First MPP Computer in China
1995



Dawning2000-I
First Cluster Computer in China
1998



Dawning2000-II
First SMP Cluster in China
1999



**Dawning3000
SUMA Cluster in China
2001**

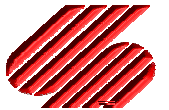


**Dawning4000-L
3Tflops Linux Cluster in China
2003**



DAWNING TC4000L SUPER COMPUTER
copyright by dawning information industry co.ltd
2003

Dawning4000-A Grid-enabling Cluster in China 2004

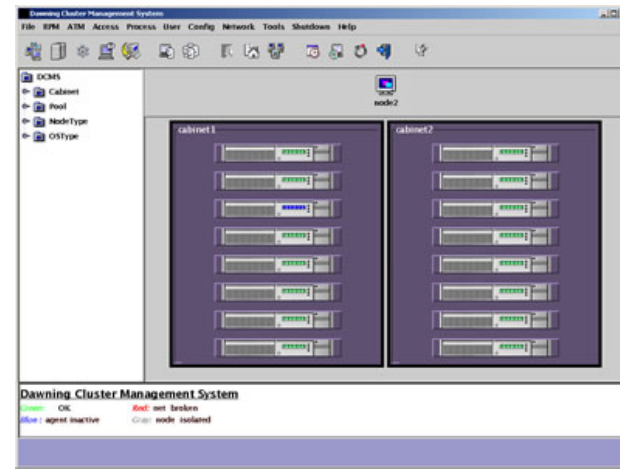


Industry Impact

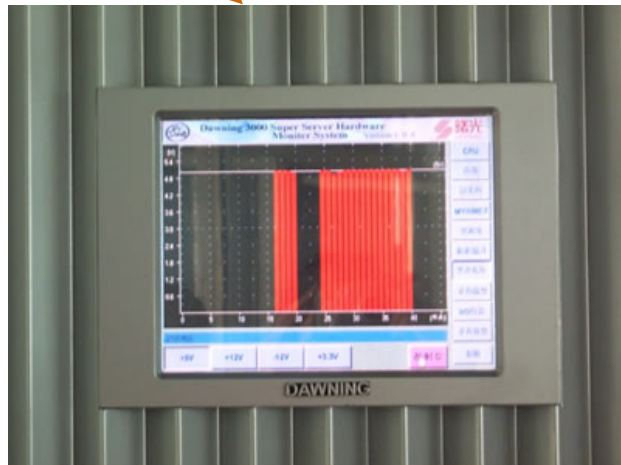
Inside Console



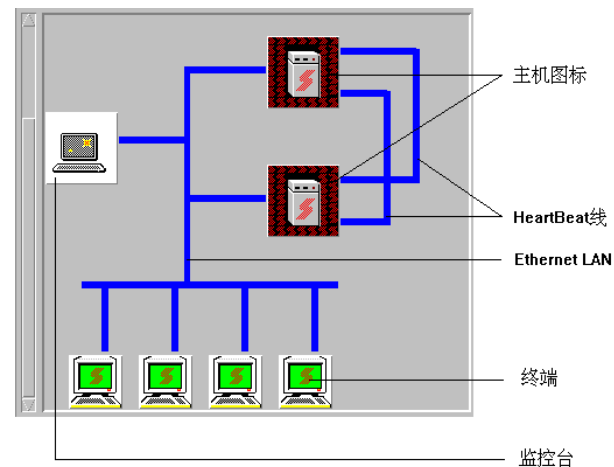
Cluster Management



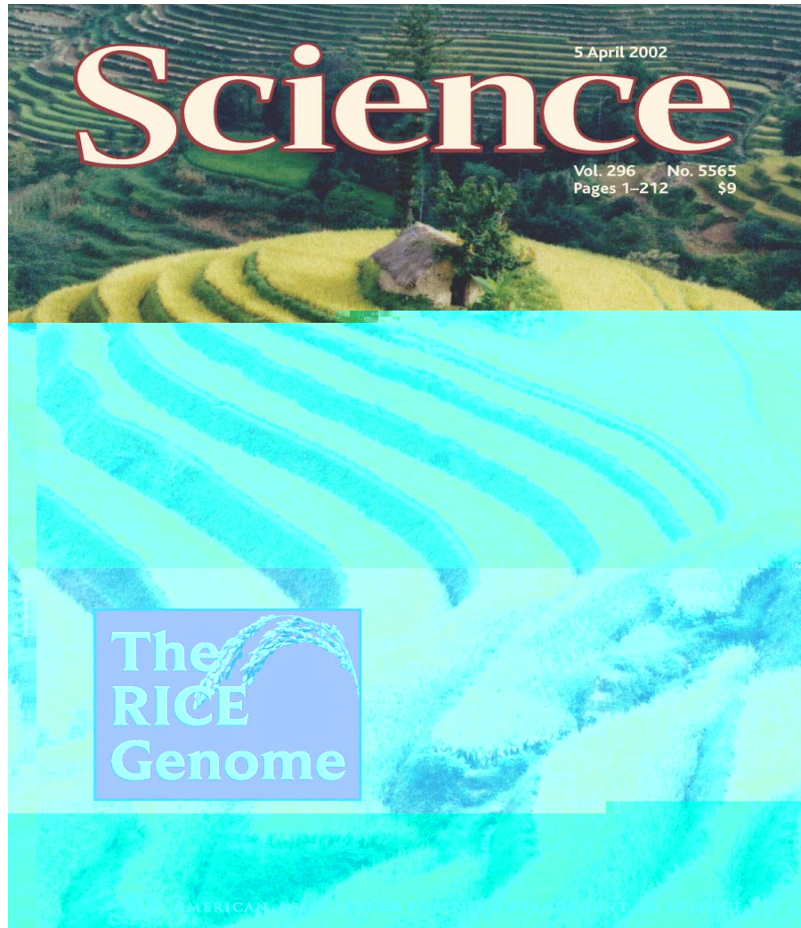
Hardware Monitor



HA Software



Draft Sequence of Rice Genome



National High Performance Computing Environment



Petroleum Industry



Changing the business rules



Dawning Server Family



Dawning4000A

CNGrid

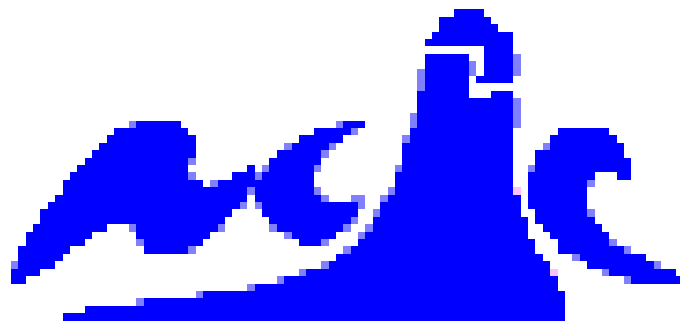
Resource-sharing



Notebook



Joint Project









Hardware

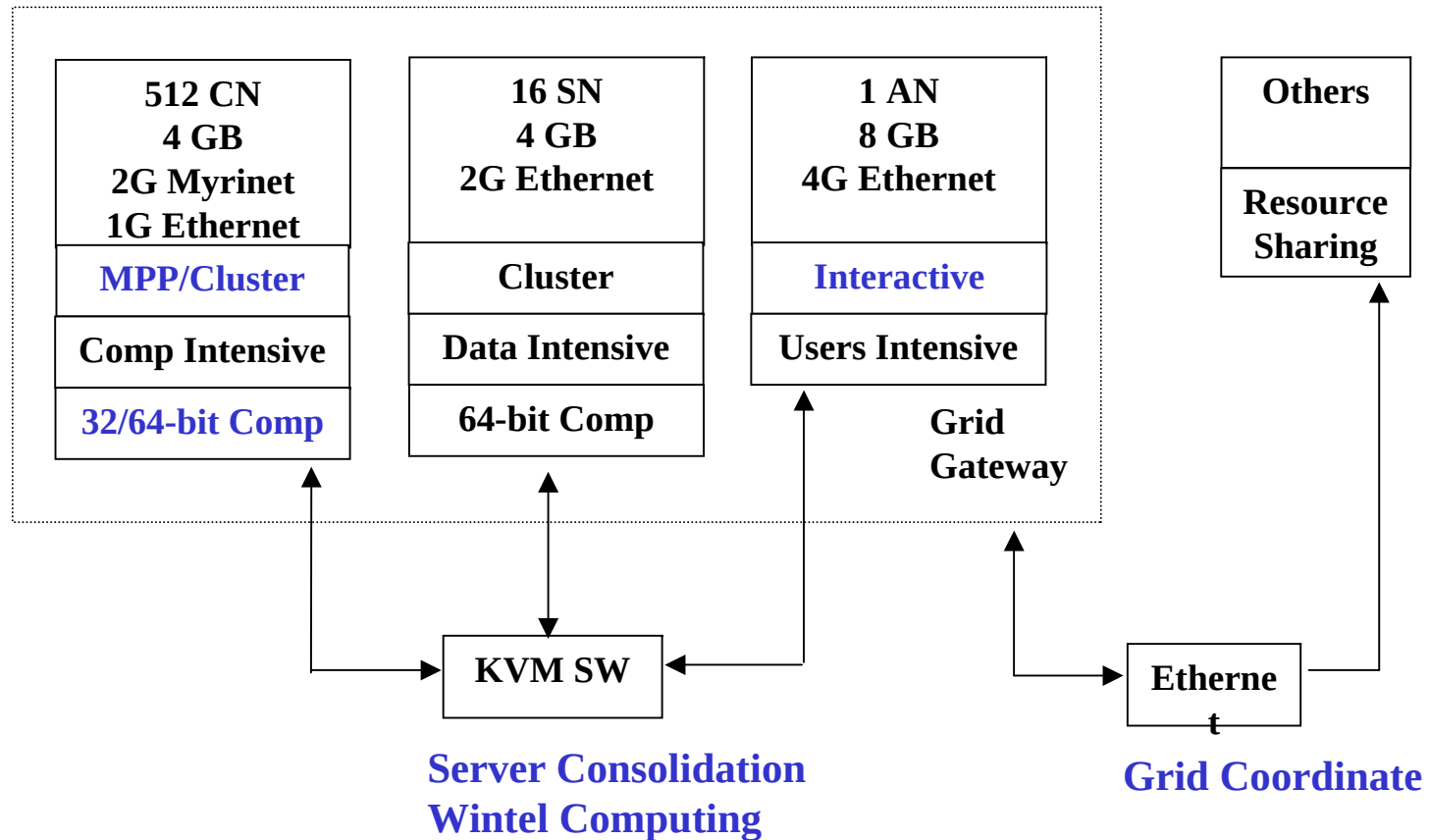
- ☐ 512 Computing Nodes (4 CPU 4GB 36GB)
- ☐ 16 Storage Nodes (overlap)
- ☐ 1 Access Node (4 CPU 8GB 36GB)
- ☐ 16 RAIDs 1.3TB
- ☐ 4 Ethernet Switches
- ☐ 12 Myrinet Switches, 512 Adapters
- ☐ 2 Management Switch 512 Management Cards
- ☐ 4 Management Consoles
- ☐ 40 Cabinets
- ☐ 1 Grid Router
- ☐ 4 Grid Keys
- ☐ 1 System Monitor







Application Models



Industry Standard Cluster

Industry Standard Architecture

- ☐ Xeon/Opteron/Itanium
- ☐ SRAM/DDR
- ☐ HyperTransport/PCI Express
- ☐ SCSI/FC/SATA/iSCSI
- ☐ Myrinet/Quatric/Infiniband/Ethernet
- ☐ Chassis/Cabinet/Cable
- ☐ Linux/GNU Compiler/Java
- ☐ GM/VIA/Verb/MPI/PVM/uDAPL
- ☐ MKL/ESSL/ACML/Scalapack/Gauss
- ☐ CMS/PBS/LSF/Luster
- ☐ Paradiam/LS-DYNA/Cerius/MM5/Blast/Oracle RAC
- ☐ LinuxNetworx/Scali



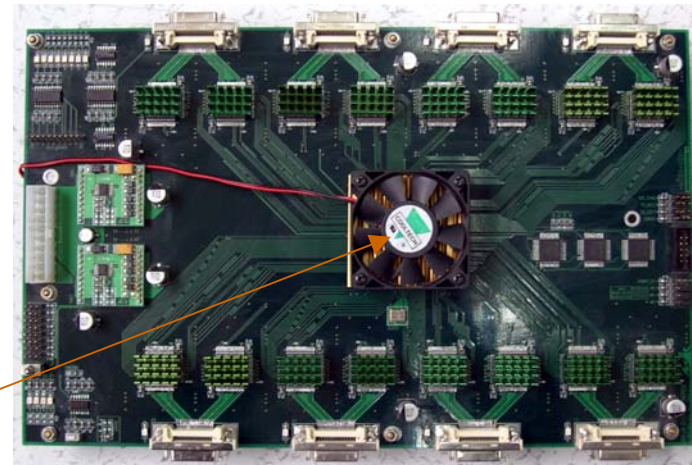
Cluster Technology

- ☐ Cluster Switch
- ☐ Parallel Environment
 - ☐ BCL4
 - ☐ DMPI/DPVM
 - ☐ JobManager
 - ☐ SVM
 - ☐ Debugger
- ☐ DCOS
 - ☐ CSMS
 - ☐ DCIS
 - ☐ DCMM
 - ☐ MultiTerm
- ☐ DCFS
- ☐ Clustone



DCNet

- ❑ 5Gbps Switch
- ❑ 8x8 Crossbar
- ❑ 250 ns switching
- ❑ Up to 1024 Nodes
- ❑ Xscale
Communication
Processor



UX8 Chip



Basic Communication Layer

BCL-1

- ☐ Kernel level communication protocol for Dawning2000-I

BCL-2

- ☐ 0-copy user-level communication protocol for Dawning2000-II

BCL-3

- ☐ Semi-user level communication for Dawning3000

BCL-4

- ☐ Parallel and Grid communication protocol for Dawning4000

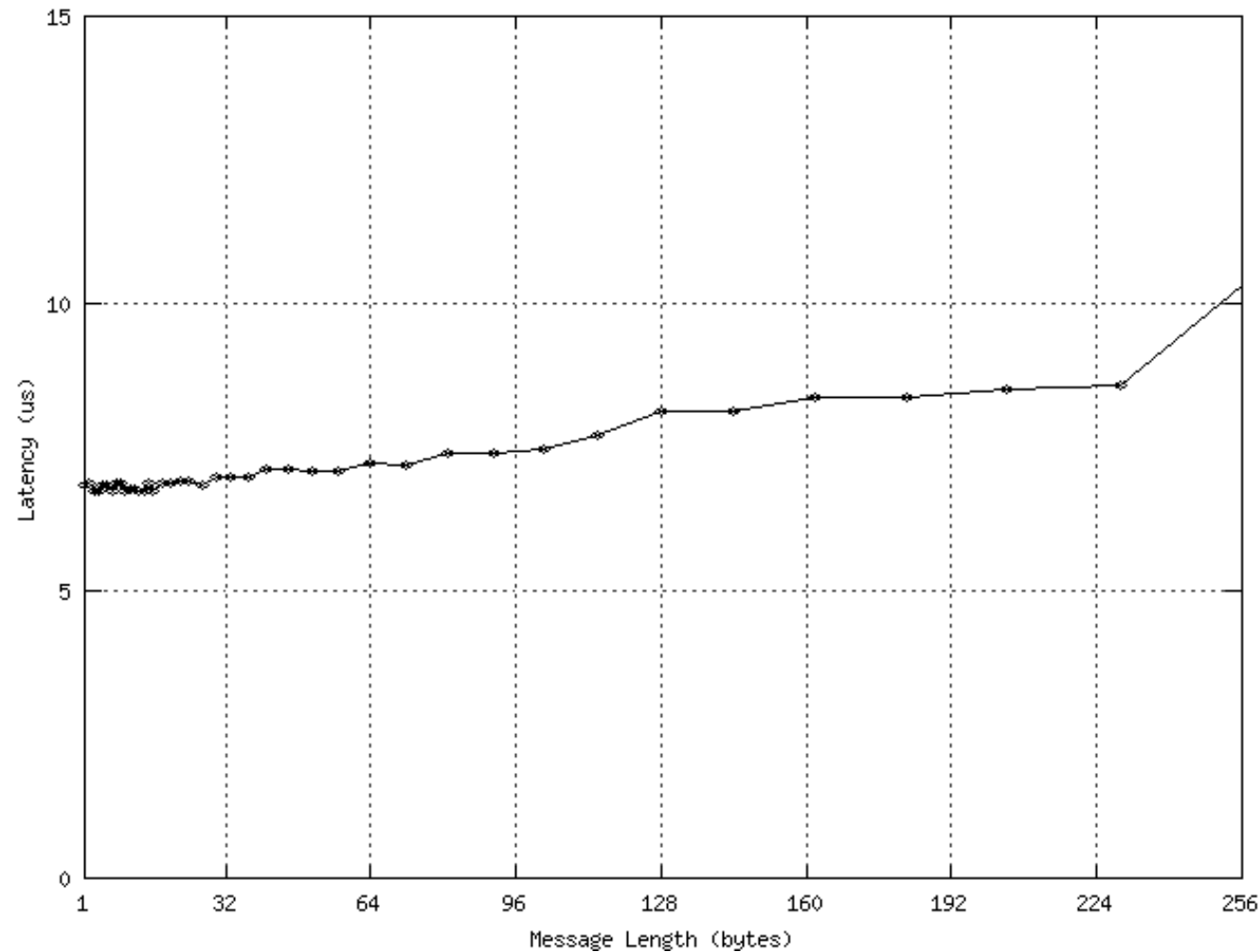


BCL4

- ☐ Based on gm 2.0.4
- ☐ Parallel communication to enable high throughput
- ☐ Support heterogeneous interconnection
- ☐ Support VIA and MPI-2 Interfaces
- ☐ Special support for inter-cluster communication to enable Grid computing
- ☐ Special support for high performance socket



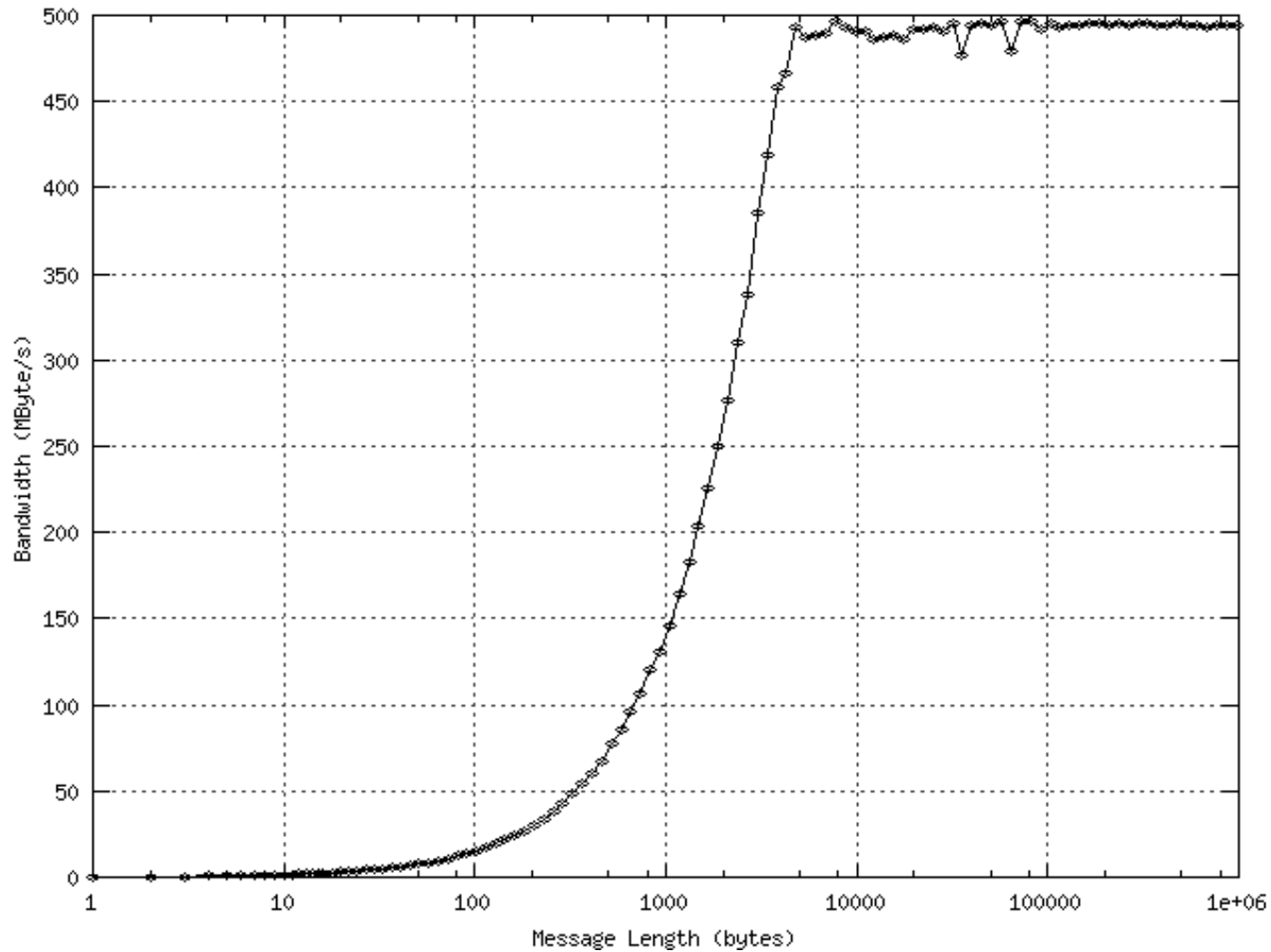
BCL-4 Latency



1.8GHz AMD Opteron
Myricom M3-E32/M3F-PCIXD-2



BCL-4 Bandwidth



1.8GHz AMD Opteron
Myricom M3-E32/M3F-PCIXD-2



DCFS

- ☐ Shared global file system
- ☐ Standard system call interface
- ☐ Multiple storage servers
- ☐ Multiple metadata servers
- ☐ High scalability
- ☐ High aggregate I/O bandwidth
- ☐ High metadata performance
- ☐ Easy management



Clustone

☐ Computation Stack

- ☐ VIA/GM/BCL
- ☐ MPI/PVM/OpenMP
- ☐ C/Fortran
- ☐ Debugger

☐ Commercial Computing

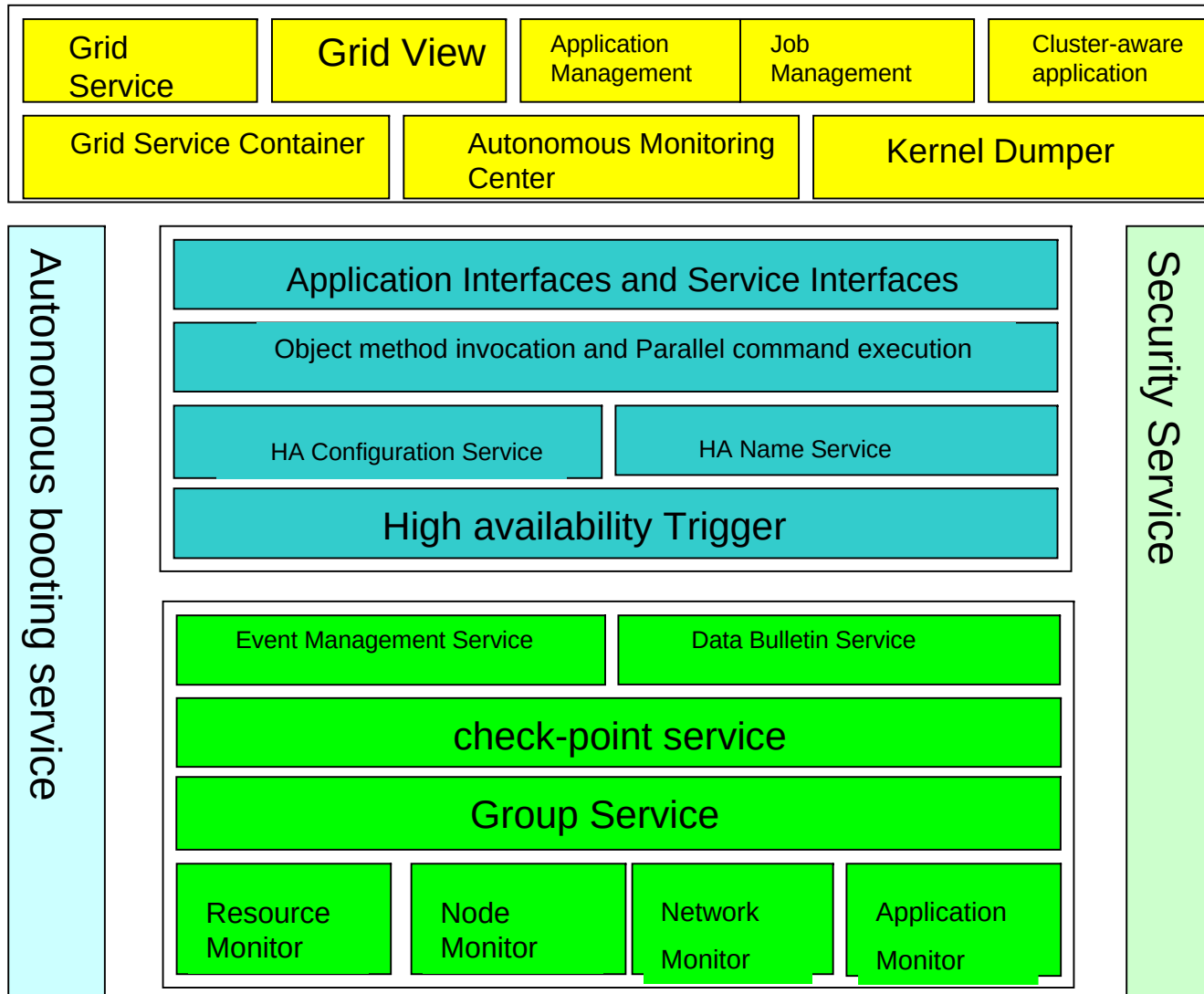
- ☐ Load Dispatcher
- ☐ Cluster Web
- ☐ HACMP/Wolfpack
- ☐ Oracle RAC

☐ Kernel of ClusterOS

- ☐ Component Framework
- ☐ Detector
- ☐ DataBase Service
- ☐ Checkpoint Service
- ☐ HA Service
- ☐ Membership
- ☐ ClusterORB



Cluster OS Hierarchy



System
software and
commercial
software

Cluster
Components

OS fault-
tolerant Kernel



Massively Cluster Computer



Features

Cluster

- ☐ Following Moore's Law
- ☐ Volume COTS
- ☐ Easily Porting
- ☐ Desktop Image
- ☐ Switch Interconnection

MPP

- ☐ Petaflops Scalability
- ☐ Sustained Performance
- ☐ Reliability
- ☐ Easily Management
- ☐ Single system image



Achieve to PetaFlops

☐ 4-Years Gap

- ☐ 2004: 10Tflops (Dawning4000A)

- ☐ 2000: 12.8Tflops (IBM ASCI White)

☐ 5-Times Gap

- ☐ Dawning4000: up to 20Tflops (Industry Standard Cluster)

- ☐ IBM ASCI Purple: 100Tflops (Specific Technology)

☐ Initiative Effort



Motherboard

- ☐ Designed by Dawning
- ☐ Manufactured by Tyan
- ☐ Heat Dissipation Simulation by AVC
- ☐ SWTX footprint (13"x16")
- ☐ Dedicated Management Slot
- ☐ Integrated KVM Switch
- ☐ 4P2U High-density Node



VIA USB2.0

Brocom Dual GE

ATI VGA

Dawning Slot

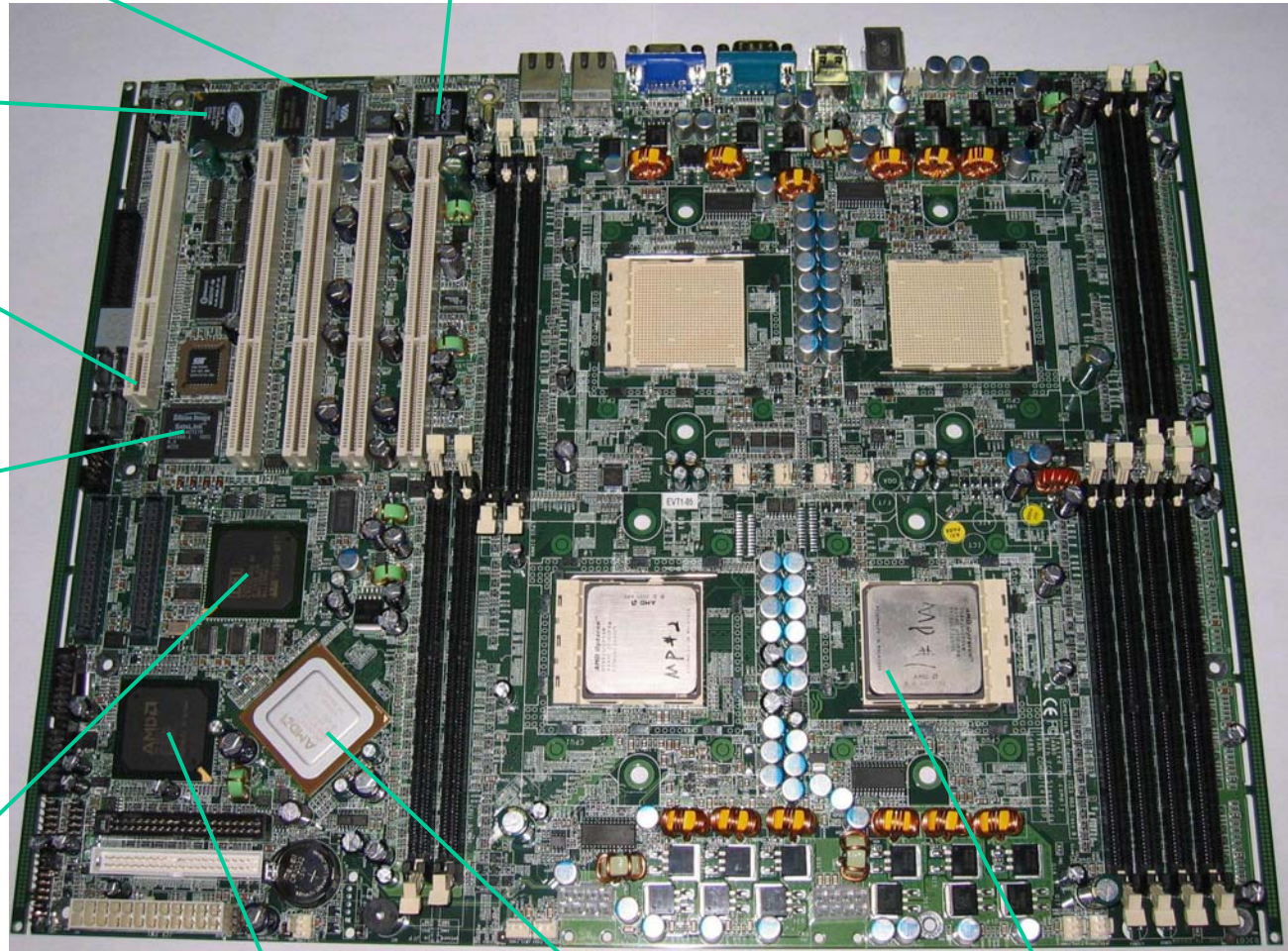
Silicon SATA

LSI Dual U3 SCSI

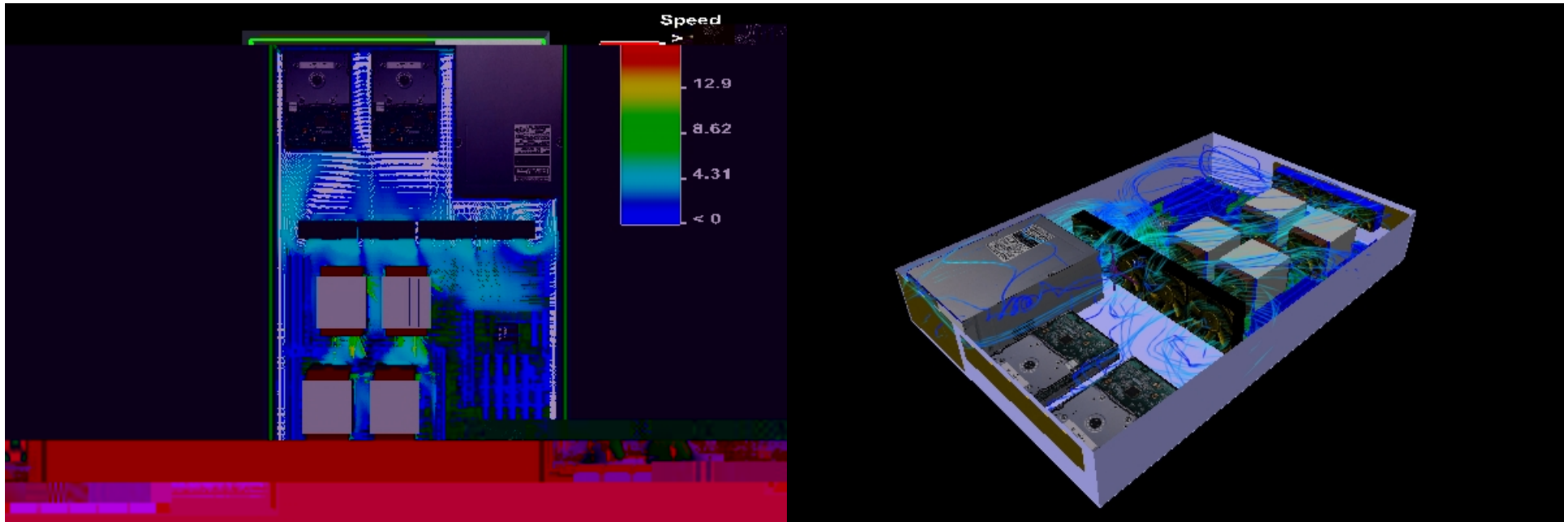
AMD 8111

AMD 8131

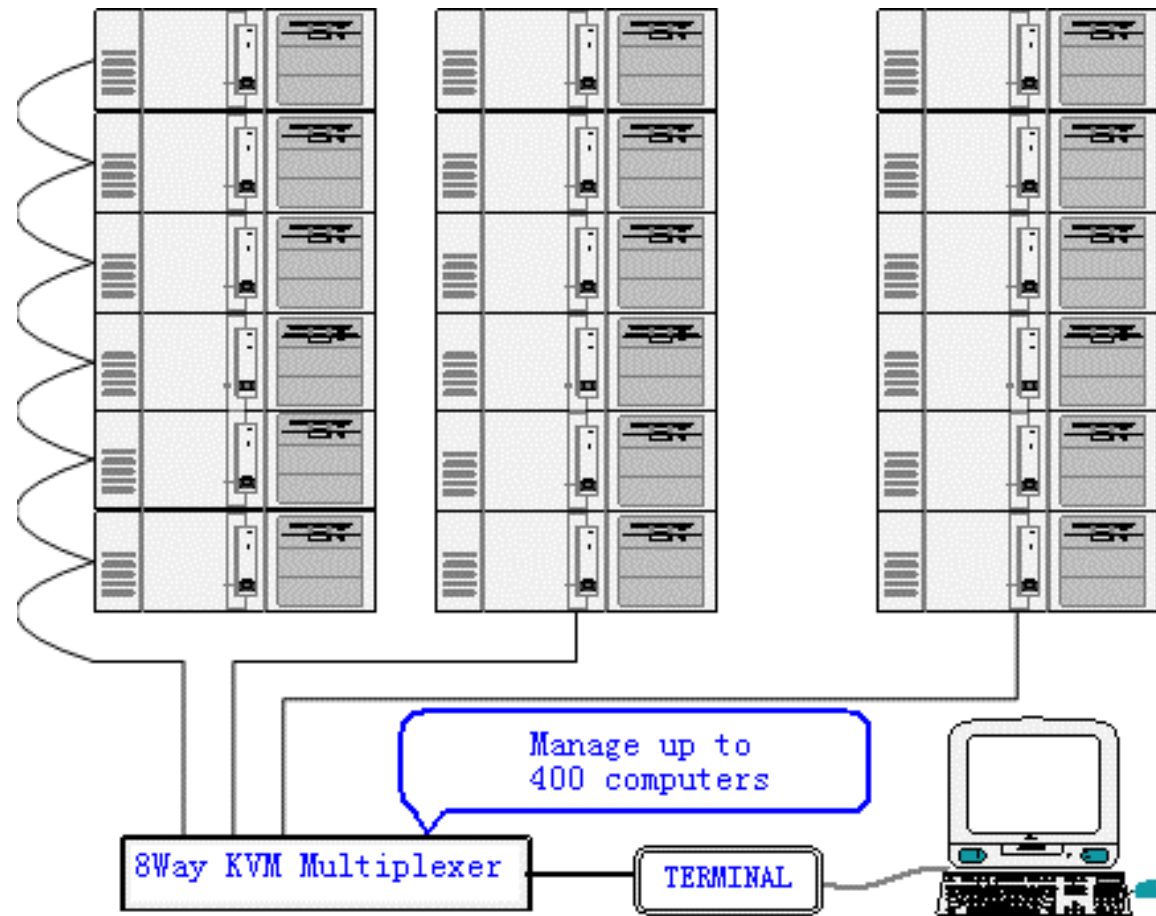
AMD Opteron



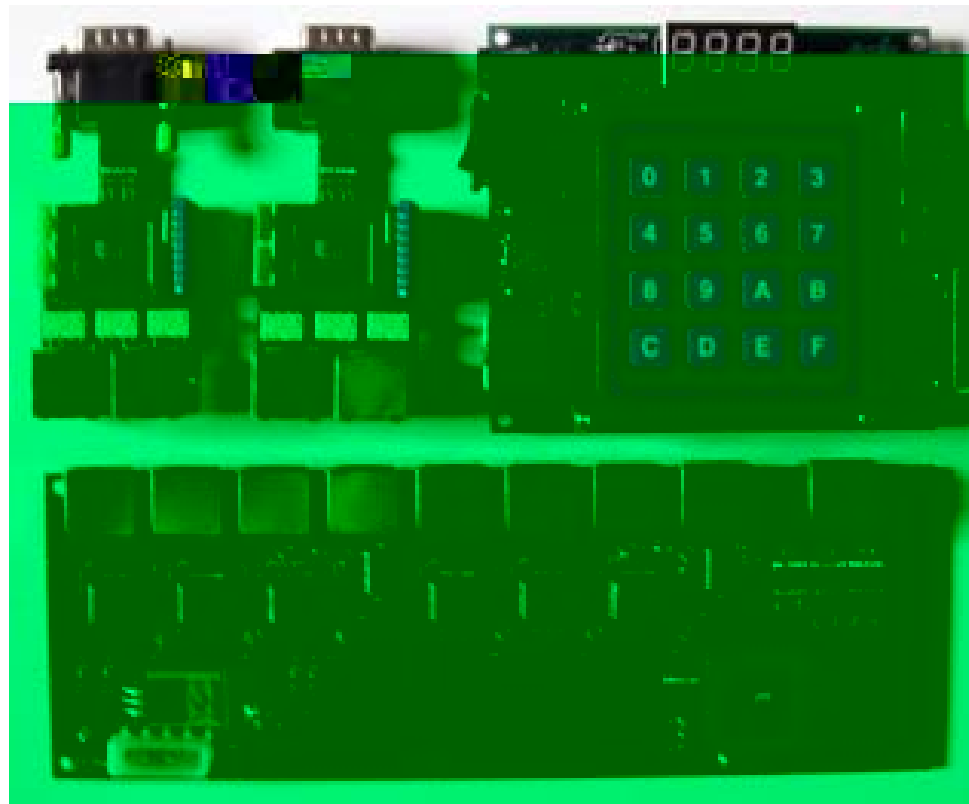
2U4P



Standalone Solution



KVM Switch









MCC LWK

- ❑ Target: improve the saturated performance of computing-intensive application
- ❑ Light Weight Kernel Linux-32
 - ❑ Hardware CPU Memory KVM Ethernet SCSI320
 - ❑ Function System call Multi-process Virtual Memory Ext2 TCP/IP NFS
- ❑ Optimization
 - ❑ Multi-User/Single-Job
 - ❑ Static Schedule of Process gang NUMA
 - ❑ Simplified MMU NUMA Memory Mapping
 - ❑ No Swap, Dynamic Library, Device Management
 - ❑ Cut out: OS Services, Daemons
 - ❑ Customized High Performance Communication
 - ❑ MPI I/O of Cluster File System



MCC Loader

- ❑ **Target: Dynamically loading multiple OSs from remote image servers for large scale cluster**
- ❑ **Key Techniques**
 - ❑ **Net Booting**
 - ❑ **Environment Building**
 - ❑ **BIOS Setup**
 - ❑ **Multicasting**
 - ❑ **Image Management**



Spreader

- ❑ **Target Management of large scale files**
- ❑ **Features**
 - ❑ **Database of metadata**
 - ❑ **Logical View**
 - ❑ **Parallel file operations**
 - ❑ **Dynamic update**



AutoAdministrator

- ❑ Target: Improve the availability of system
- ❑ Features
 - ❑ Memory Leak
 - ❑ System Overload Protection
 - ❑ Cluster Software Aging
 - ❑ Node Crash
- ❑ Technique
 - ❑ Failure Check
 - ❑ Dynamic Prediction
 - ❑ Recovery
 - ❑ Intelligent Administration Policy





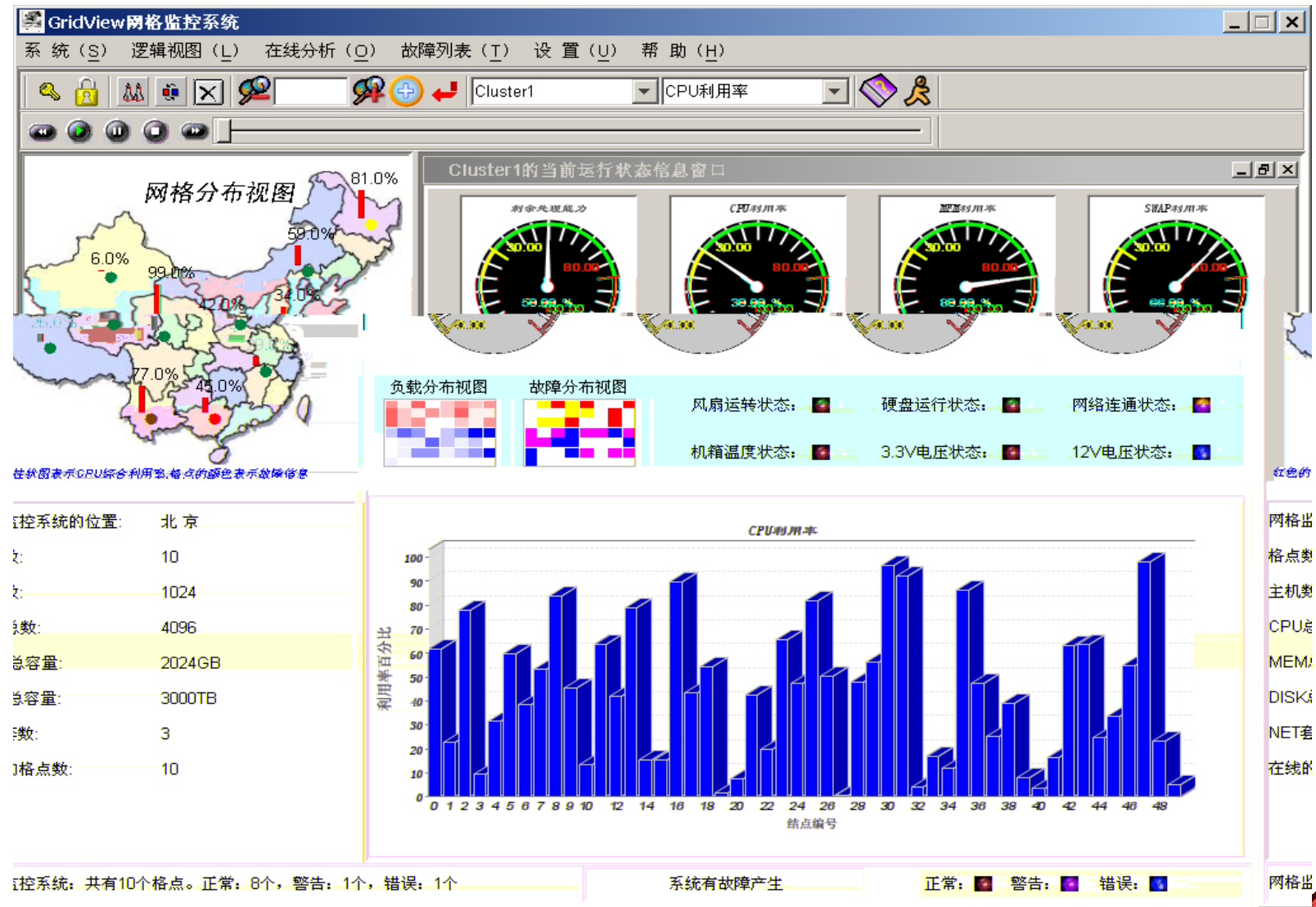
Grid-enabling Components

- ☐ **GridView**
- ☐ **Grid Router**
- ☐ **Grid Key**
- ☐ **Grid Gateway**
- ☐ **Grid I/O**
- ☐ **GridOS**

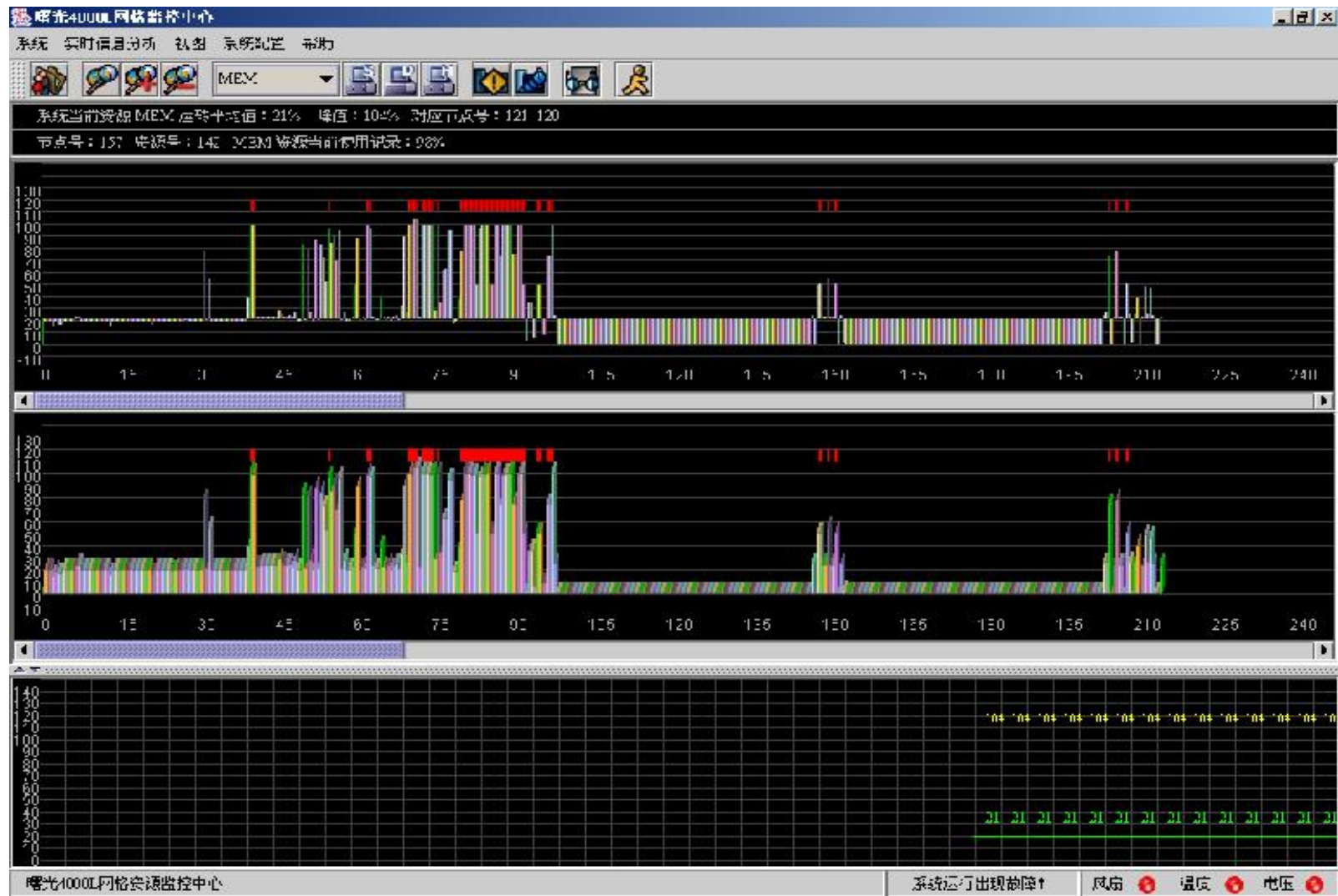




Grid View



Snapshot





Grid Router

- ☐ Router between System and Grid Environment
- ☐ 2Gbps Smart Adapter
- ☐ Remote Management
- ☐ Access Internet Resource
 - ☐ Parallel NAT
- ☐ System Interconnection in Grid
 - ☐ Parallel VPN
- ☐ Access Internal Service
 - ☐ Load Balancing
 - ☐ Full-life Authentication
 - ☐ Parallel Firewall
 - ☐ Parallel Ftp, Telnet, X-win Proxy
- ☐ High Availability



Grid Key

☐ A USB Security Device on Client

☐ Features

- Access resources in Grid
- Authentication
- Service Setup
- Digital Signature
- Account





Grid I/O

- ❑ Grid ftp
- ❑ Grid MPI-IO
- ❑ VegaFS



Vega Grid

GOS Version 1.0 (Alpha)

[Linux](#) (~ 44M) [Win32](#) (~ 43M)

[User Guide](#) [API Manual](#)

<http://vega.ict.ac.cn>



THANKS